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Section 1: Introduction

1.1 Traffic Management Manual for Work on Roadways

Traffic Management	The strategies designed to safely mitigate the impact of construction, rehabilitation, maintenance, incident management and special events on roadways to maintain mobility and worker safety. The documentation of strategies is completed using a Traffic Management Plan.
Traffic Control	The effective use of temporary traffic control devices to protect workers and move road users safely through a work zone. Traffic Control is implemented using a Traffic Management Plan

The Traffic Management Manual for Work on Roadways (TMM or the Manual)—including the updates that may be issued from time to time—is required for planning and implementing traffic control for work zones on Provincial highways.

The Manual updates and combines information from these previous Ministry publications:

- 2015 Interim Traffic Management Manual for Work on Roadways
- Traffic Management Guidelines for Work on Roadways, 2001
- Traffic Control Manual for Work on Roadways, 1999
- Technical Circulars relevant to temporary traffic control (up to publication date)

1.1.1 Purpose of the Manual

This Manual outlines fundamental principles and guidelines for traffic management and traffic control in work zones with the goal of ensuring the protection of workers and the safe and efficient movement of road users through the work zone.

It also includes standards for designing, applying, positioning, installing, maintaining, and inspecting various types of temporary traffic control devices through work zones, including signs, channelizing devices, signals, barricades, markings, lighting, and others.

1.1.2 Minimum Standards in the Manual

The standards in this Manual are mandatory for Provincial roadways.

Minimum standards are prescribed in the Manual for controlling traffic through highway work zones. Several typical situations are illustrated to show the recommended application of standard devices for planned, scheduled work on roadways.

All projects involving highway or street construction or maintenance, utility work, and incident management shall control road users—drivers, pedestrians, cyclists, and those with disabilities—and provide them with the information and guidance they need to successfully traverse the work zone.

Complying with minimum standards is mandatory for all works on Provincial roadways, including but not limited to, work performed by contractors, sub-contractors, and public utilities.

The standards are designed to minimize impacts on existing traffic, and maintain vehicle, pedestrian, cyclist, and worker safety during road projects. They apply to all sizes of roadway projects.

If minimum standards cannot be met—for whatever reason—the contractor shall discuss the matter with the Road Authority to develop a suitable resolution. It is understood that it may not be possible to meet these minimum standards in emergency situations.

For Ministry Roadways, Ministry Maintenance Contractors may be exempt from standards in this Manual, as defined in their Highway Maintenance Agreement.

1.1.3 Applying the Principles in the Manual

1. The term “work zone” in the Manual (see also *Appendix A: Glossary*) refers to the area which extends from the first traffic control device to the last traffic control device as seen by the travelling public, installed specific to the work.
2. The Manual provides guiding principles for traffic management and traffic control in work zones. Only typical situations are illustrated because various situations may be encountered in practice. Users may benefit from using the principles and examples provided in the Manual to develop customized Traffic Control Plans.
3. The traffic management and control requirements described and illustrated in the Manual are generally the minimum required. Additional measures may be necessary in certain circumstances and under particular conditions.
4. For situations not specifically addressed in this Manual, traffic control procedures should be established by appropriately modifying the general traffic control principles specified in the Manual.
5. No single standard sequence of signs or devices can be used as a fixed arrangement for all conditions and locations. When the Manual indicates that something shall be done, there may be circumstances where strict compliance with the requirement is not reasonable, and it will be necessary to deviate from the requirement. In these cases, written justification for the modification must be recorded and depending on the complexity of the modification, written permission may need to be granted by the Road Authority.
6. The principles provided in the Manual are applicable to both urban and rural areas.
7. Traffic conditions on urban streets are characterized by lower speeds, widely ranging traffic volumes, limited maneuvering space, frequent turns and cross-movements, significant pedestrian movement, and other obstructions. Customization of traffic control layouts is often necessary to maintain access and mobility.
8. Traffic conditions on rural highways are characterized by higher speeds and fewer access points. Advance warning is critical to provide sufficient response time in rural Traffic Control Plans.
9. The Manual provides some information about Ministry contracts and project cycles. Other Road Authorities may find it useful to modify these systems and tools for their own projects.

1.1.4 Organization and Content of the Manual

PART A – TRAFFIC MANAGEMENT

Key traffic management principles and requirements, and processes for developing consistent Traffic Management Plans for the Ministry. Other Road Authorities may use the information for reference, modifying it to suit their own requirements.

Section 2: Fundamentals of Traffic Management and Control

Key traffic management and traffic control principles and requirements, and practical processes for the consistent development of Traffic Management Plans for the Ministry.

Section 3: Traffic Management Plans

Information about Ministry project categories and the planning requirements for each category, the inspection, maintenance, and documentation of traffic control, and instructions for developing project-specific Traffic Management Plans, each of which will have one or more of these components:

- Traffic Control Plan
- Incident Management Plan
- Public Information Plan
- Implementation Plan

PART B – TRAFFIC CONTROL

Basic principles and minimum standards for the design, application, installation, and maintenance of traffic control through work zones.

Section 4: Temporary Traffic Control Devices

Information about temporary traffic control devices (signs, signals, lighting, channelizing devices, pavement markings, etc.) and the standards for their use and placement.

Section 5: Traffic Control Persons

Information about Traffic Control Persons, including their responsibilities, conduct, apparel, equipment, communications, and safety.

Sections 6 to 19: Traffic Control Layouts

General instructions, descriptions, standards, guidance, options, and diagrams regarding traffic control layouts for typical roadway projects.

APPENDICES

Appendix A: Glossary

Definitions of terms and acronyms used in this manual.

Appendix B: Standard Construction Signs

Illustrations, sizes, and applications of standard construction signs.

Appendix C: Templates for Traffic Management and Traffic Control Plans

Templates for developing Traffic Management Plans and Traffic Control Plans for Category 1, 2, and 3 projects.

Appendix D: Traffic Management Plan Audit Forms

Ministry forms for auditing a Traffic Management Plan both before the project commences and when it is under way.

Appendix E: Lane Closure Request Form

Sample of the Ministry's Work Notification/Lane Closure Request and Approval Form and link to the online form.

Appendix F: Tables A to D

Tables A to D repeated in one location for quick reference and printing.

Appendix G: Pilot Car Load Movement Guidelines

Guidelines for pilot cars used in the movement of oversize and overweight loads.

1.1.5 Using the Manual

The Manual—including any updates—shall be used for planning and implementing traffic control for work on Provincial highways.

- **Part A - Traffic Management** (Sections 2 and 3) is used for work on Provincial roadways and may be used for work on municipal roadways.
- **Part B - Traffic Control** (Sections 4 to 19) is mandatory for work on Provincial roadways and may be used for work on municipal roadways.

All users of this manual should have a thorough understanding of its contents, including the definitions found in *Appendix A: Glossary*.

All provisions for driver, pedestrian, cyclist, and worker protection established in this Manual shall be implemented by:

- The Ministry and its contractors
- Public utilities and their contractors
- Others who have authorization to work on, or adjacent to, Provincial roadways

The Manual has several features that may help users to find the information they need:

1. Sections are numbered 1 through 19, and Appendices are labelled A through F.
2. Sections and subsections are numbered using a decimal system whereby a notation like 2.5.7 refers to Section 2, subsection 5, sub-subsection 7. Each section and subsection deals with one topic or one aspect of a larger topic.
3. Cross-references to information within the Manual are shown in italics (e.g., see *Section 5.1.1: Traffic Control Supervisor*), whereas other publication titles are not italicized. Hence, every italicized reference resides within the Manual.
4. Tables A, B, C, and D are shown and referenced in Sections 6 to 19, and collected together in Appendix F for quick access.
5. To find a topic in a printed Manual, use the Manual Table of Contents at the beginning of the Manual or the individual Section Contents at the beginning of each section.
6. To find a topic in the digital Manual, search for key words or phrases, or use the Manual Table of Contents or individual Section Contents. To navigate to cited websites, click on the links.

1.1.6 Updating the Manual

1. Insert new and revised pages into printed copies of the Manual as soon as they are available. Remove and discard obsolete pages.
2. The insertion of new and revised pages should be recorded in the Manual Update Record on the page following the cover page at the beginning of the Manual.
3. Users are encouraged to contribute to this process by submitting suggestions for corrections and new content to the Director of Traffic and Safety Engineering, Ministry of Transportation and Infrastructure at MoTITMM@gov.bc.ca.

1.2 Road Authority and Prime Contractor Responsibilities

1.2.1 Jurisdiction

Provincial legislation and municipal bylaws authorize Road Authorities to control traffic for work.

- Federal roadways, such as highways through National Parks, fall under the jurisdiction of the Federal Government.
- The Ministry is the Road Authority for, and has jurisdiction over, all Provincial highways, as well as roads outside incorporated areas.
- In municipalities, the Road Authority is the municipal government, which has jurisdiction over the entire incorporated area except for the Provincial highways.

1.2.2 Work Permits

The Road Authority shall authorize the work or provide a work permit before any work zone or lane closure may be established. Road Authorities may grant "continuing" permits for frequent activities, such as those undertaken by utility providers.

See *Appendix E: Lane Closure Request Form* for a sample of the Ministry's Work Notification/Lane Closure Request and Approval Form and a link to the online form, which can be populated online and printed for submission to the Ministry.

1.2.3 Traffic Control Responsibilities

The Road Authority shall clearly assign traffic control responsibility in writing through a permit, contract, or protocol agreement to ensure that:

- All supervisors and crews are thoroughly familiar with and trained in the applicable safe work practices.
- Immediate and decisive action is taken when safe and approved work methods are not followed.
- Each crew member wears the required personal safety equipment and apparel when working on or crossing the highway.
- Traffic control is provided where necessary by using temporary traffic control devices and/or traffic control persons, as outlined in this Manual.

The responsibilities identified on the following pages shall be assigned and implemented to ensure a successful Traffic Management Plan. The assignment of specific individuals to specific roles will vary with the size and complexity of the work.

1. Road Authority

For contracted works, the Road Authority identifies hazards, provides known information to the Prime Contractor, issues permits or authorization to commence work, periodically audits traffic control setups on the roadway, and monitors traffic issues.

1. If the Road Authority is supervising and directing work being completed by in-house staff, the Road Authority is also the Prime Contractor.
2. If the Road Authority is using a Consulting Services contract where traffic control is to be managed by the Consultant, then a Notice of Assignment should be completed, designating the Consultant as the Prime Contractor.

2. Prime Contractor

For all works on Provincial highways, the Ministry designates an individual or organization as the Prime Contractor—through a signed agreement, such as the Notice of Assignment—and the Prime Contractor is responsible for traffic management and control.

Many utility companies operate on Ministry right-of-way (electrical, telecommunications, gas, etc.). Work carried out by utility companies on Ministry right-of-way is tied directly to the conditions specified in the permit, including the H1080 – Work Notification/Lane Closure Request and Approval form (see *Appendix E*). The permit and its associated conditions that specify requirements for utility operations on Ministry right-of-way are issued by the Ministry of Transportation and Infrastructure's local District office where the utility company operates.

For the purposes of this Manual, the Prime Contractor is the organization directly constructing or maintaining works on a Provincial highway and is responsible for:

1. Obtaining Ministry authorization to work on the roadway;
2. Developing an acceptable Traffic Management Plan; and
3. Implementing the Traffic Management Plan in accordance with Ministry requirements.

3. Management

Traffic control activities associated with work should have a management structure for these activities to ensure that all supervisors and workers are thoroughly familiar with, and trained in, the applicable safe work practices.

Management and site supervision personnel shall monitor the effectiveness of traffic control, take immediate and decisive action when safe and approved work methods are not followed, and document and make necessary changes as issues arise.

Individuals assigned traffic control responsibilities shall have adequate knowledge and training in all facets of traffic control, including:

- traffic control operations (those conducting traffic control shall be trained in a manner acceptable to WorkSafeBC)
- Part 18 of WorkSafeBC's Occupational Health and Safety Regulation
- the content of this Manual, other relevant publications and technical circulars, and manuals and regulations that govern other jurisdictions (when and where required by particular projects)

4. Site Supervisor/Foreman/Superintendent

The Site Supervisor/Foreman/Superintendent shall ensure that:

- Each crew member is familiar with the Traffic Control Plan.
- Each crew member wears the required safety apparel and uses the required equipment when working on or crossing the highway.
- The work area is protected by implementing the Traffic Control Plan which uses various signs, channelizing devices, and other temporary traffic control devices and measures described in this Manual.

The Site Supervisor/Foreman/Superintendent cannot be a Traffic Control Manager, Traffic Control Supervisor, or Traffic Control Person for the project unless the dual functions are authorized by the Road Authority in the contract or permit.

5. Traffic Engineer

The Traffic Engineer is a Professional Engineer who is licensed in British Columbia and qualified and experienced in traffic management planning and highway safety. They are a requirement for Category 3 Traffic Management Plans, and may be required, as specified by the Road Authority, on Category 2 Traffic Management Plans.

Traffic Engineers may also sign-off in cases where minimum standards cannot be met. In all cases, the matter shall be discussed with the Road Authority and documented.

6. Traffic Control Manager

The Traffic Control Manager is appointed by the Prime Contractor, and is responsible for preparing, implementing, and managing the Prime Contractor's Traffic Management Plan and sub-plans. This includes reviewing, evaluating, and approving the details in the Traffic Control Plan, including the traffic control layouts.

These are the typical duties and responsibilities of the Traffic Control Manager:

- fully implements the Traffic Control Plan
- monitors traffic operations to determine the effectiveness of the Traffic Control Plan
- ensures that the Traffic Management Plan remains current
- oversees modifications to the Traffic Management Plan as required by changes to the construction schedule, accommodation of special events, and changes to sub-plans
- ensures that daily traffic control logs are maintained
- exercises full line authority over all Traffic Control Persons on the work site
- finalizes traffic control measures with the Prime Contractor's Traffic Engineer where these are required by the Road Authority or the standards in this Manual
- sets up and implements a monitoring schedule for both active and inactive work periods throughout the course of the project
- directs the Prime Contractor's Incident Management Plan
- directs the Prime Contractor's Public Information Plan
- directs the Prime Contractor's Implementation Plan

- attends regular meetings with the Road Authority Representative on behalf of the Prime Contractor to discuss project performance, issues, and plans

The role of Traffic Control Manager should be assigned only to a qualified person who is knowledgeable about traffic management principles and requirements, and who has suitable traffic management work experience or training. The Traffic Control Manager may be an employee of the Prime Contractor or a sub-contractor to the Prime Contractor, and may be the designated Traffic Control Supervisor or another qualified person.

7. Traffic Control Supervisors and Traffic Control Persons

For the roles and responsibilities associated with these positions, see *Section 5.1.1: Traffic Control Supervisor* and *Section 5.1.2: Traffic Control Persons (TCPs).*

1.2.4 Road Work Authorization

The Road Authority authorizes the Prime Contractor to work on a Provincial (or municipal) roadway where the work will impact road users. There is typically conditions (requirements) associated with the authorization which may be outlined in a:

- **Contract:** Traffic management requirements may form part of the contract conditions.
- **Permit to Construct Works upon Highways/Roadways:** Traffic management requirements may form part of the permit conditions.
- **Lane Closure Request/ Acceptance:** The Ministry's Work Notification/Lane Closure Request and Approval Form may be useful where traffic is impacted but no physical works are being constructed that require a permit (see *Appendix E: Lane Closure Request Form).*
- **Letter of Authorization:** May be used when no standard form suits the circumstances. Care should be taken to ensure that the appropriate requirements and conditions are communicated to the Prime Contractor in writing when authorizing work on a highway.

1.2.5 Plan Submission and Review Process

The Prime Contractor shall plan and document the Traffic Management that will be implemented based on the requirements set by the Road Authority. No work may be performed on the roadway unless the Prime Contractor's Traffic Management Plan meets the requirements and has been reviewed and accepted by the Road Authority.

Section 2.5: Establishing Work on Roadways outlines the submission and review process in detail used by the Prime Contractor.

PART A – TRAFFIC MANAGEMENT

Section 2: Fundamentals of Traffic Management and Traffic Control

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PART A – TRAFFIC MANAGEMENT

Section 2: Fundamentals of Traffic Management and Traffic Control

2.1 Principles of Traffic Management

2.1.1 Traffic Management Goals and Objectives

The goal of effective traffic management is to **protect workers** and **accommodate road users** while work is underway on roadways. Temporary closures or peak-period capacity reductions are not acceptable on many roadways, and even minor traffic disruptions may have efficiency and safety implications.

Identifying potential impacts and addressing them in the planning process will assist with the establishment of project-specific traffic management objectives and appropriate traffic control, and these in turn will help to ensure successful work.



Figure 2.1: A Work Zone

Effective traffic management ensures that:

- Project and site-specific issues are accommodated.
- Potential traffic impacts are reviewed and mitigated.
- Stakeholders are notified of potential impacts.
- Evaluation and modification occurs as needed.

When planning a project, it is important to use a **clear, consistent, and structured** process for developing the Traffic Management Plan so traffic issues are carefully considered and the Plan is reviewed for suitability. This Manual provides this process for Prime Contractors, Ministry staff, and other Road Authorities.

2.1.2 Traffic Management Principles

Four primary questions should be addressed when considering traffic management for a project:

1. What type of work is occurring?
2. What is the likelihood of traffic delays?
3. What special circumstances must be accommodated?
4. What type of traffic control is required for the work?

2.1.3 Traffic Management Process

The process for planning and evaluating traffic management on every project should consider the following:

1. Identification of the work.
2. Clear definition of roles and responsibilities.
3. Development of clear and consistent Traffic Management Plans appropriate to the complexity of the work and the site.
4. Regular and effective evaluation of Traffic Management Plans.
5. Management and documentation of ongoing changes.

As illustrated in *Figure 2.2: Traffic Management Process Life Cycle,* traffic management issues should be considered during the project design phase to identify the appropriate project category and develop the contract.

The Ministry's internal process traffic management plan acceptance for major works projects is outlined in *Figure 2.3: Ministry TMP Acceptance Process for Major Works – Category 1 and 2* and *Figure 2.4: Ministry TMP Acceptance Process for Major Works – Category 3*.

A Traffic Management Plan is a dynamic document. Once work is underway, performance should be monitored, and plans modified to reflect changing traffic management issues, construction schedules, and deadlines.

Once the project is completed, the effectiveness of the Traffic Management Plan should be assessed, and the lessons learned incorporated into the planning for future projects.

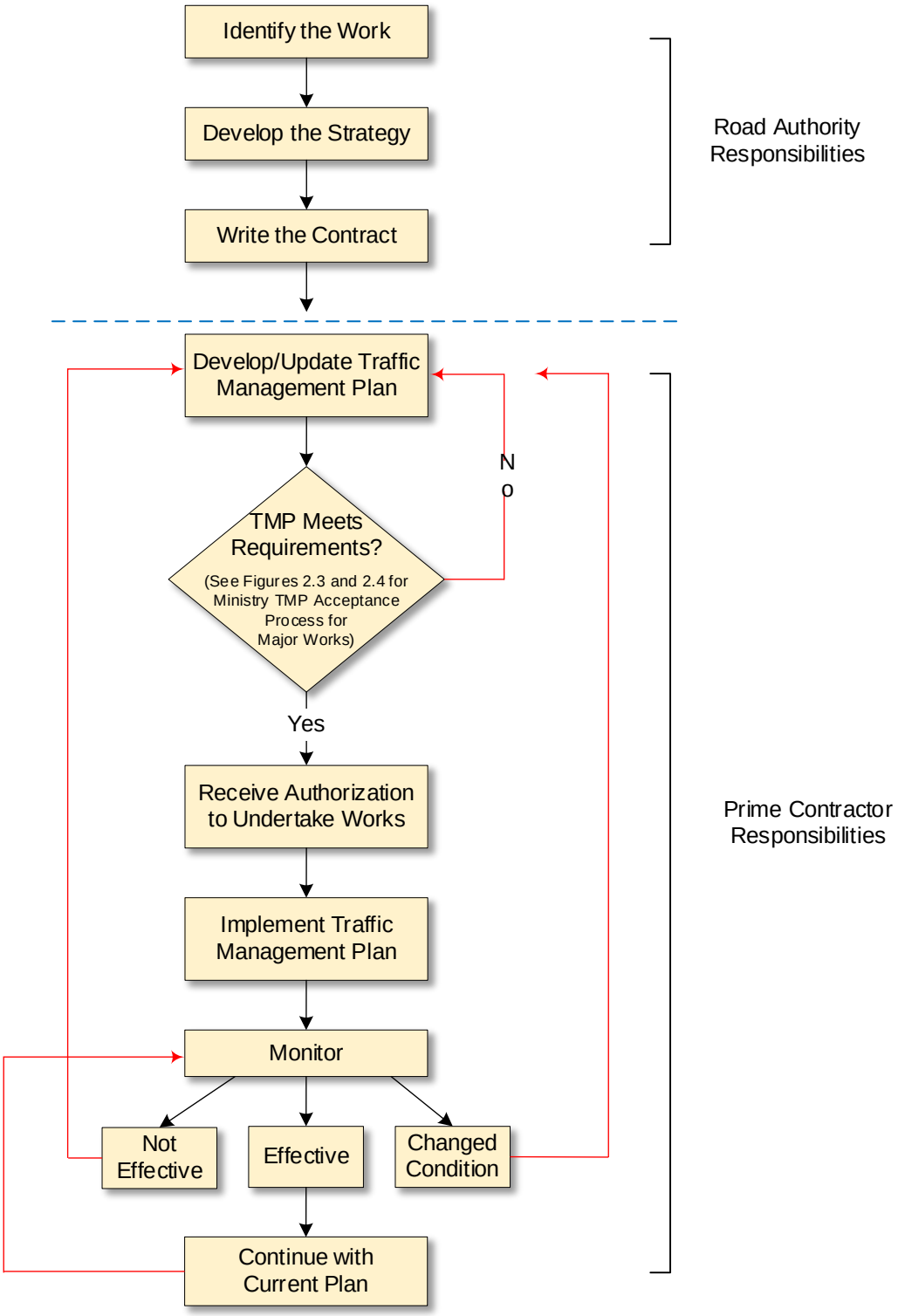


Figure 2.2: Traffic Management Process Life Cycle

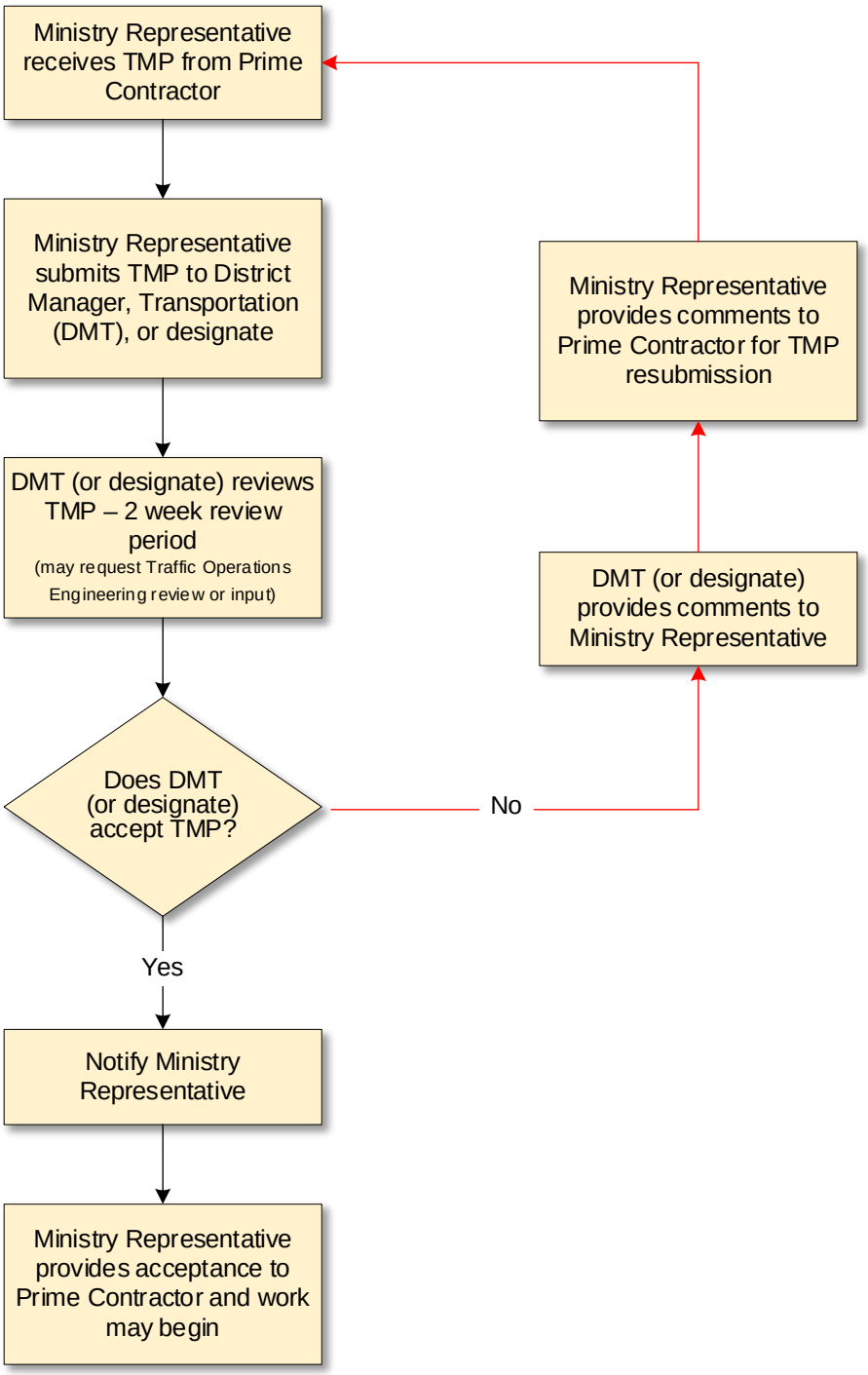


Figure 2.3: Ministry TMP Acceptance Process for Major Works – Category 1 and 2

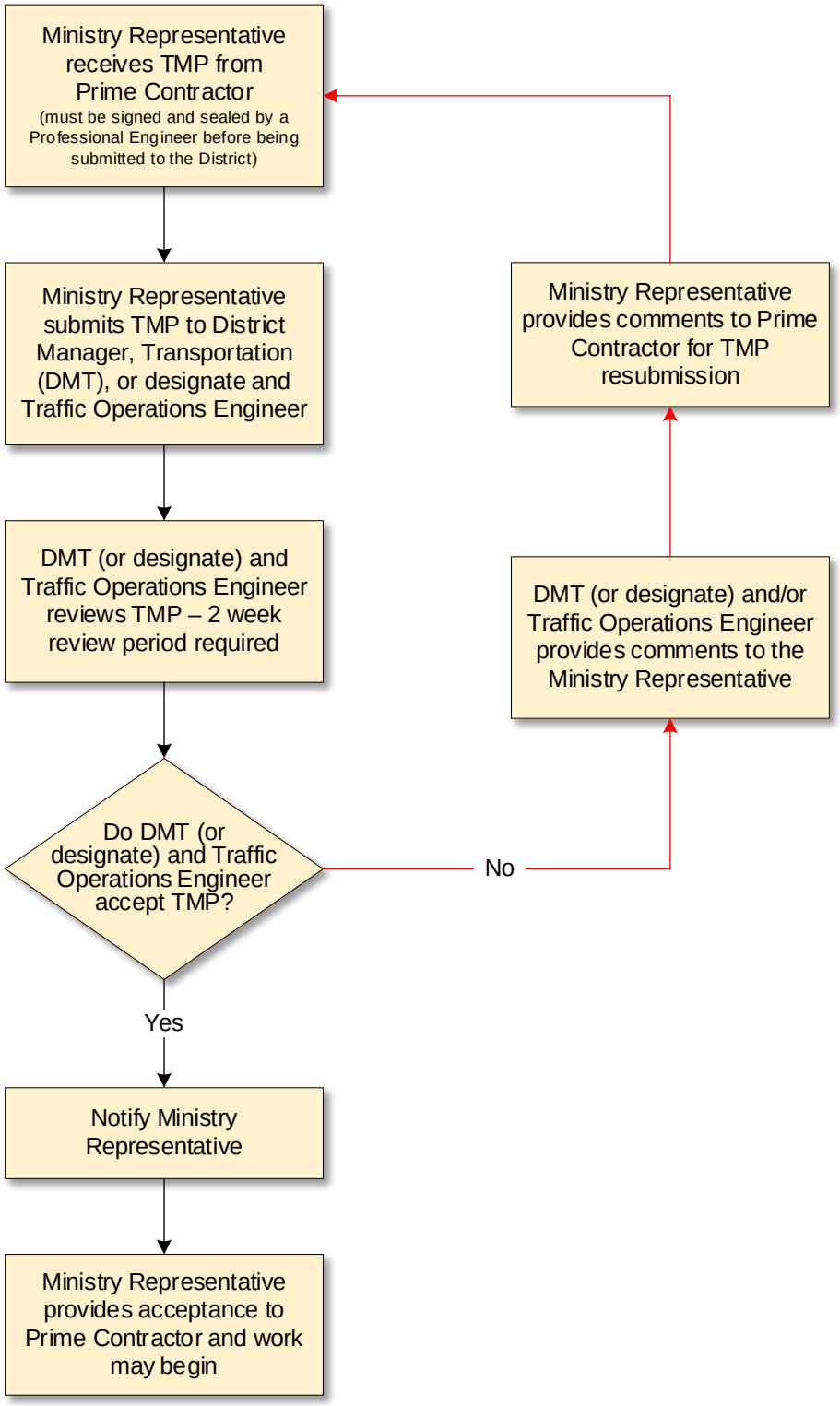


Figure 2.4: Ministry TMP Acceptance Process for Major Works – Category 3

2.2 Principles of Traffic Control

Traffic Control	The implementation of the Traffic Management Plan through the effective use of temporary traffic control devices to protect workers and move road users safely through the work zone.

2.2.1 Hierarchy of Controls

Use the hierarchy of controls below to manage interaction between road users and the work zone. Most work zones will require a combination of these controls.

1. Elimination Controls

Completely eliminate public traffic from the work area and construction traffic through constructed detours, alternate routes, or otherwise isolating the work zone.

2. Engineering Controls

Use prescribed temporary traffic control devices and layouts to effectively control traffic through a work zone, permitting public traffic and construction to interact without the use of Traffic Control Persons.

3. Administrative Controls

Schedule the work at times when traffic volume are low, reducing the exposure of public traffic to construction.

4. Traffic Control Persons

Use Traffic Control Persons where the strategies described above have been considered and deemed unsuitable to effectively manage traffic.

2.2.2 Planning and Implementation by Trained Persons

Traffic control is planned, designed, approved, selected, installed, operated and maintained by trained persons. Complex field layouts require comprehensive plans and the services of an experienced traffic specialist.

Training is usually obtained through course work, programs, and experience in the field of Traffic Control or Traffic Engineering. Traffic Control Persons shall retain evidence of traffic control training that is, at minimum, acceptable to WorkSafeBC.

2.2.3 Minimum Disruption to Road Users

Work on roadways can cause inconvenience and significant disruption for road users. Road user movement and time through a work zone should be inhibited as little as reasonably possible, while addressing all known hazards.

1. Driving conditions through work zones should be as similar as possible to normal driving conditions.
2. Scheduling of work should consider congestion and delay.
3. Keep reduced construction speed limits reasonable to the given hazards and road conditions.
4. Provide notification to all stakeholders of the work.
    - Various media, including newspaper, radio, internet.
    - Businesses and residences for which access will be affected are provided with sufficient advance notice to plan alternative actions or routes.
    - Emergency services are considered and accommodated.
    - Railroad, transit, and commercial vehicle use is accommodated or alternative routing is planned.

2.2.4 Clear, Positive Direction for Road Users

The measures implemented for traffic control should be designed to provide clear, positive direction, as if every road user were approaching the area for the first time.

1. Drivers should be provided with positive guidance in advance of and through work zones by means of temporary traffic control devices that are effective under varying conditions of light and weather.
2. Channelization of traffic should be accomplished by using pavement markings, signing, flexible posts or drums, tubular markers, cones, delineators, barricades, and other light-weight devices.
3. Pavement markings should be modified as needed for long-duration work.
4. When the use of temporary traffic control devices is inadequate for clear direction, Traffic Control Persons should be considered.
5. When traffic control devices are not needed, they shall be removed, covered, or turned.

2.2.5 Accommodation of Other Road Users

All road users should be accommodated through the work zone. Typical traffic control is directed towards drivers of passenger and commercial vehicles.

Motorcycles

Road surface conditions can impact motorcycle operation more so than other vehicles. This is particularly associated with resurfacing projects involving milling and paving.

Transit and School Buses

Discussions with the local Transit Authority or School District should discuss the impacts of the work and address the location of alternative passenger pick-up and drop-off points, if required. It can be expected that groups of bus users can congregate within the work zone. Accommodations may be required for storage and protection of these users.

Emergency Vehicles

Emergency vehicles shall be accommodated through the work zone at any time. This includes volunteer responders (i.e. fire fighters) who may be in personal vehicles.

Police Enforcement

Where there is a need for enhanced police enforcement in a work zone, consideration should be given to providing safe areas where vehicles can park and where enforcement officers can pull vehicles over.

Pedestrians

Provisions should be made to accommodate pedestrians through work zones. Temporary pedestrian facilities should include accessibility features consistent with those present in the existing facility.

Mobility-assisted pedestrians, wheelchairs, walkers, or scooters require smooth and solid transition and running surfaces. Care should be taken to prevent tipping.

Visually-impaired pedestrians may require some additional guidance. Barricades should be used to block off closed areas and a distinguishable “path” of tubular markers should be erected for the pedestrians to follow. Assistance may also be provided by a Traffic Control Persons or designate.

Cyclists

Provisions should be made to accommodate cyclists through work zones. Consideration should be made to the road surface and work zone or detour length.

Under the provisions of the Motor Vehicle Act, cyclists have the same rights and duties as the operators of motor vehicles so the signage and traffic control measures used for drivers also apply to cyclists.

For more information, see *Section 18: Traffic Control Layouts – Bicycle Lanes.*

2.2.6 Maintenance of Traffic Control Devices

Traffic control devices shall be inspected and maintained during both active and inactive work to ensure that they are not missing, and that they are clean, properly positioned, and effective in all weather and light conditions.

The frequency of inspection will be determined by the operational needs of the project. The operational needs are directly related to factors such as:

- highway functional classification,
- traffic volumes,
- number and choice of traffic control devices deployed, and
- duration of work (active and inactive periods)

The higher the values for these factors, the more frequent the inspections, which should be documented in a Quality Control Plan.

2.2.7 Roadside Safety

Maintaining roadside safety requires constant attention during the project life cycle due to the potential for hazard changes. Traffic control for the project should be modified as required to ensure it is effective in protecting workers and road users. In addition:

- Traffic control devices shall be crash-worthy (American Association of State Highway and Transportation Officials (AASHTO) Manual for Assessing Safety Hardware (MASH) tested) and not create a greater hazard than the hazard against which they are intended to protect road users.
- A travelled roadway should always be maintained by keeping it clear of construction equipment, materials, and devices as practical. The travelled roadway and any equipment should be clearly marked and delineated as required.
- The shoulder area adjacent to the travelled lanes should be maintained as unencumbered as practical. Due to highway right-of-way constraints, the shoulder area may have to be used for the storage of construction equipment and materials.
- Clear zone requirements are not applicable in work zones.
- The storage of construction equipment and materials should be as far from the travelled lanes as possible unless site constraints require them to be placed on the shoulder. In these cases, they shall be clearly marked and delineated.
- Drop-offs and excavations require special considerations (see *Section 6.5: Treatment of Drop-Offs and Travel Lane Excavations*).

Elements of roadside safety are discussed throughout this manual.

2.2.8 Traffic Control during Inactive Work

Traffic control shall be maintained during periods of inactive work, including during seasonal shut downs. When a work zone is left inactive, these steps should be taken:

1. Traffic control should anticipate the most adverse conditions that could reasonably be expected to occur.
2. Store construction equipment and materials clear of the travelled roadway or mark and delineate the area around them.
3. Ensure that all temporary traffic control devices are secured against weather and vandalism.
4. Remove, cover, or turn off any temporary signs or devices that are not applicable. Reinstate any permanent devices which are now applicable.
5. Establish a schedule to monitor and maintain the site during inactive work.
6. Make available to the Road Authority the name and phone number of individuals who can be reached 24/7 to respond promptly to concerns.

2.2.9 Work near Railways

Railways are under the jurisdiction of the railway owner. Any work that may impact railway track or right-of-way requires authorization from the railway owner.

Issues for consideration include:

- the potential queuing of traffic onto railway tracks
- works affecting track operations (e.g., blasting rocks onto tracks)
- disruption of traffic signals with pre-emption operations
- Transport Canada requirements for working around railways

See *Section 7.17: Work Near a Rail System Grade Crossing* for additional guidance.

2.2.10 Work near Accesses and Driveways

It is necessary to consider access and egress from private, commercial, and public locations.

Prior to the commencement of work, discussion with business and property owners is highly recommended to facilitate a coordinated and systematic approach to managing traffic flow into and out of the businesses/residential properties. This may include:

- controlling access and egress to the properties
- providing alternative access with supplemental signage
- placing signage within the properties to assist vehicular movement
- assessing sight distance
- establishing “closure” times or planning work in the affected area to minimize disruption to customers/residents
- limiting access